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Calculation of Basketball Scoring Probabilities

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Agenda

Results of basketball scoring calculations for Steph Curry (2022-2023 Season)

1. **Overall Shooting Statistics**
   P(Make), P(Miss), proportion of 2pt vs 3pt shots
2. **Binomial Probabilities**
   P(3 makes in 4 shots), P(4 of 5 are 3-pointers)
3. **Conditional Probabilities — Future**
   P(Make|3pt), P(Lead|3pt), P(Make|2pt), P(Lead|2pt) via Bayes' Theorem
4. **Conditional Probabilities — Retrospective**
   P(3pt|Made), P(2pt|Made) via Bayes' Theorem

Overall Shooting Statistics

Dataset: 1,434 shots by Steph Curry (2022-2023 NBA Season)

Shot Results:

- P(Make) = 703 / 1,434 = 0.4902 (49.02%)
- P(Miss) = 731 / 1,434 = 0.5098 (50.98%)

Shot Type Distribution:

- P(Three-pointer) = 786 / 1,434 = 0.5481 (54.81%)
- P(Two-pointer) = 648 / 1,434 = 0.4519 (45.19%)

Key Insight:

Curry attempts more 3-pointers than 2-pointers, but his overall shooting percentage is just under 50%.

Binomial Probabilities

Q1: P(Steph makes 3 of the next 4 shots)

p = P(Make) = 0.4902, n = 4, k = 3

$P(X=3) = C(4,3) \times 0.4902^3 \times 0.5098^1$

$P(X=3) = 4 \times 0.1178 \times 0.5098 = 0.2402$ (24.02%)

Q2: P(4 of the next 5 shots are three-pointers)

p = P(3pt) = 0.5481, n = 5, k = 4

$P(X=4) = C(5,4) \times 0.5481^4 \times 0.4519^1$

$P(X=4) = 5 \times 0.0902 \times 0.4519 = 0.2039$ (20.39%)

Assumptions:

1. Each shot is an independent event (outcome of one shot does not affect the next)
2. The probability of success remains constant across shots
3. Each shot has only two outcomes: make or miss

Note: In reality, these assumptions may be violated — fatigue, defense, and momentum can affect shooting probability.

Conditional Probabilities - Future

If the next shot is a THREE-POINTER:

- **P(Make | 3pt) via Bayes' Theorem:**
  = P(3pt|Make) × P(Make) / P(3pt)
  = (329/703) × (703/1434) / (786/1434)
  = 0.4680 × 0.4902 / 0.5481 = 0.4186 (41.86%)

- **P(Lead | 3pt) via Bayes' Theorem:**
  = P(3pt|Lead) × P(Lead) / P(3pt)
  = (388/722) × (722/1434) / (786/1434)
  = 0.5374 × 0.5035 / 0.5481 = 0.4936 (49.36%)

- **If the next shot is a TWO-POINTER:**

- **P(Make | 2pt) = P(2pt|Make) × P(Make) / P(2pt)**
  = 0.5320 × 0.4902 / 0.4519 = 0.5772 (57.72%)

- **P(Lead | 2pt) = P(2pt|Lead) × P(Lead) / P(2pt)**
  = 0.4626 × 0.5035 / 0.4519 = 0.5154 (51.54%)

Conditional Probabilities - Retrospective

If Steph just MADE a shot, what type was it?

- **P(Three-pointer | Made) via Bayes' Theorem:**
  Step 1: Define notation
  P(Made|3pt) = 329/786 = 0.4186
  P(3pt) = 786/1434 = 0.5481
  P(Made) = 703/1434 = 0.4902
  Step 2: Apply Bayes' Theorem
  P(3pt|Made) = P(Made|3pt) × P(3pt) / P(Made)
  = 0.4186 × 0.5481 / 0.4902 = 0.4680 (46.80%)
  Verification: 329 / 703 = 0.4680 ✓

- **P(Two-pointer | Made) via Bayes' Theorem:**
  P(2pt|Made) = P(Made|2pt) × P(2pt) / P(Made)
  = 0.5772 × 0.4519 / 0.4902 = 0.5320 (53.20%)
  Verification: 374 / 703 = 0.5320 ✓

- **Check: P(3pt|Made) + P(2pt|Made) = 0.4680 + 0.5320 = 1.0000 ✓**